

Problem

An autotransformer with a 40 percent tap is supplied by an 880-V, 60-Hz source and is used for stepdown operation. A 5-kVA load operating at unity power factor is connected to the secondary terminals. Find:

(a) the secondary voltage,

(b) the secondary current,

(c) the primary current.

Step-by-step solution

Step 1 of 3

(a)

Consider that an autotransformer is supplied by an 880 V, 60 Hz source.

Assume that the total number of turns in the transformer are N . Therefore,

$$N_1 + N_2 = N$$

Consider that the transformer is used for step-down operation.

Consider the following expression for the voltage ratio of the step-down autotransformer:

$$\frac{V_1}{V_2} = \frac{N_1 + N_2}{N_2}$$

Consider that 40 percent tap is supplied by the source. Therefore,

$$\begin{aligned} N_2 &= (40\%)N \\ &= 0.4N \end{aligned}$$

Therefore,

$$\frac{V_1}{V_2} = \frac{N}{0.4N}$$

$$\frac{V_1}{V_2} = \frac{1}{0.4}$$

$$V_2 = 0.4V_1$$

Substitute 880 V for V_1 .

$$\begin{aligned} V_2 &= 0.4(880 \text{ V}) \\ &= 352 \text{ V} \end{aligned}$$

Therefore, the secondary voltage V_2 is 352 V.

Step 2 of 3

(b)

The power in the load is,

$$P = S \cos \phi$$

The secondary of the transformer is connected to a 5 kVA load operating at unity power factor.

Substitute 5 kVA for **S**, 1 for **cos φ**.

$$\begin{aligned} P &= (5 \text{ kVA})(1) \\ &= 5 \text{ kW} \end{aligned}$$

The expression for the load power is,

$$P = V_2 I_2$$

Therefore,

$$I_2 = \frac{P}{V_2}$$

Substitute 5 kW for P, and 352 V for **V₂**.

$$\begin{aligned} I_2 &= \frac{5 \text{ kW}}{352 \text{ V}} \\ &= \frac{5000}{352} \\ &= 14.2 \text{ A} \end{aligned}$$

Thus, the secondary current **I₂** is **14.2 A**.

Step 3 of 3

(c)

Consider the following expression for the current ratio of the step-down autotransformer:

$$\begin{aligned} \frac{I_1}{I_2} &= \frac{N_2}{N_1 + N_2} \\ &= \frac{0.4N}{N} \\ &= 0.4 \end{aligned}$$

Therefore,

$$I_1 = 0.4 I_2$$

Substitute 14.2 A for **I₂**.

$$\begin{aligned} I_1 &= 0.4(14.2 \text{ A}) \\ &= 5.68 \text{ A} \end{aligned}$$

Thus, the primary current **I₁** is **5.68 A**.